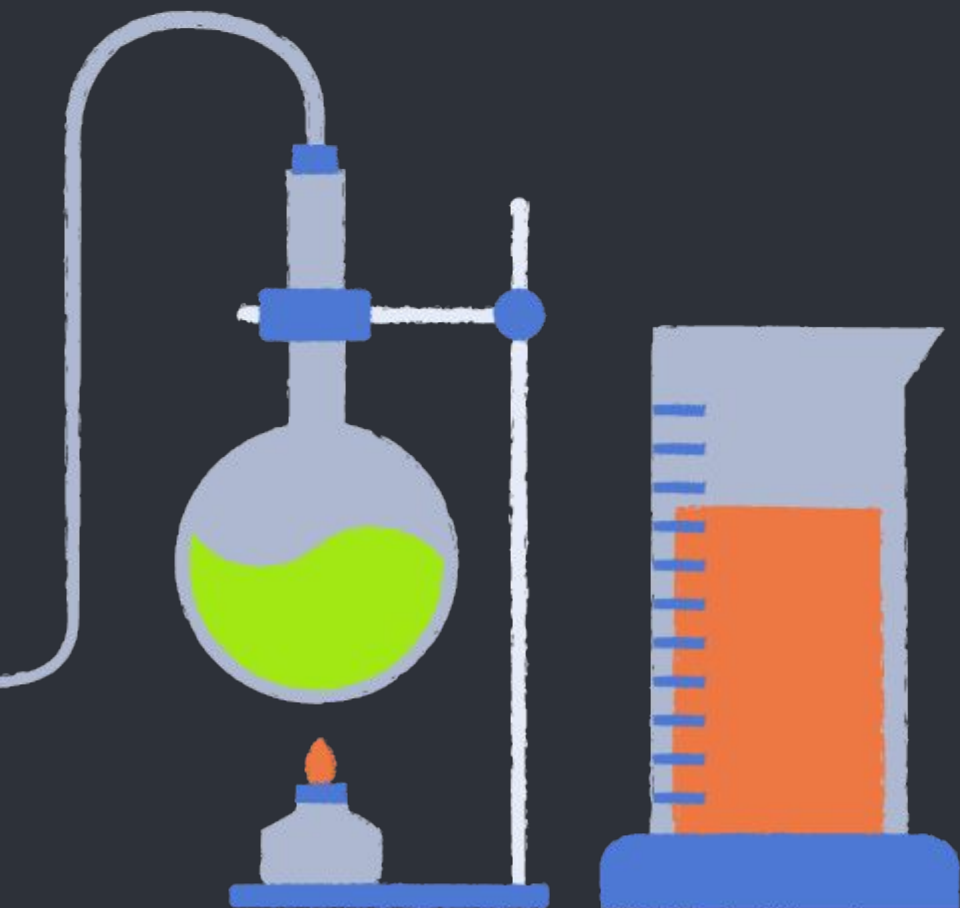


Class 10th One -Shot series

Heredity

BIOLOGY





Topics To Be Covered

- Heredity
- Father of Genetics
- Some Important Terms
- What did Mendel do?
- Law of dominance, segregation and independent assortment
- How traits are expressed
- Sex Determination

Variation



When we observe a corn field we see that there is little variation among the individual plants.



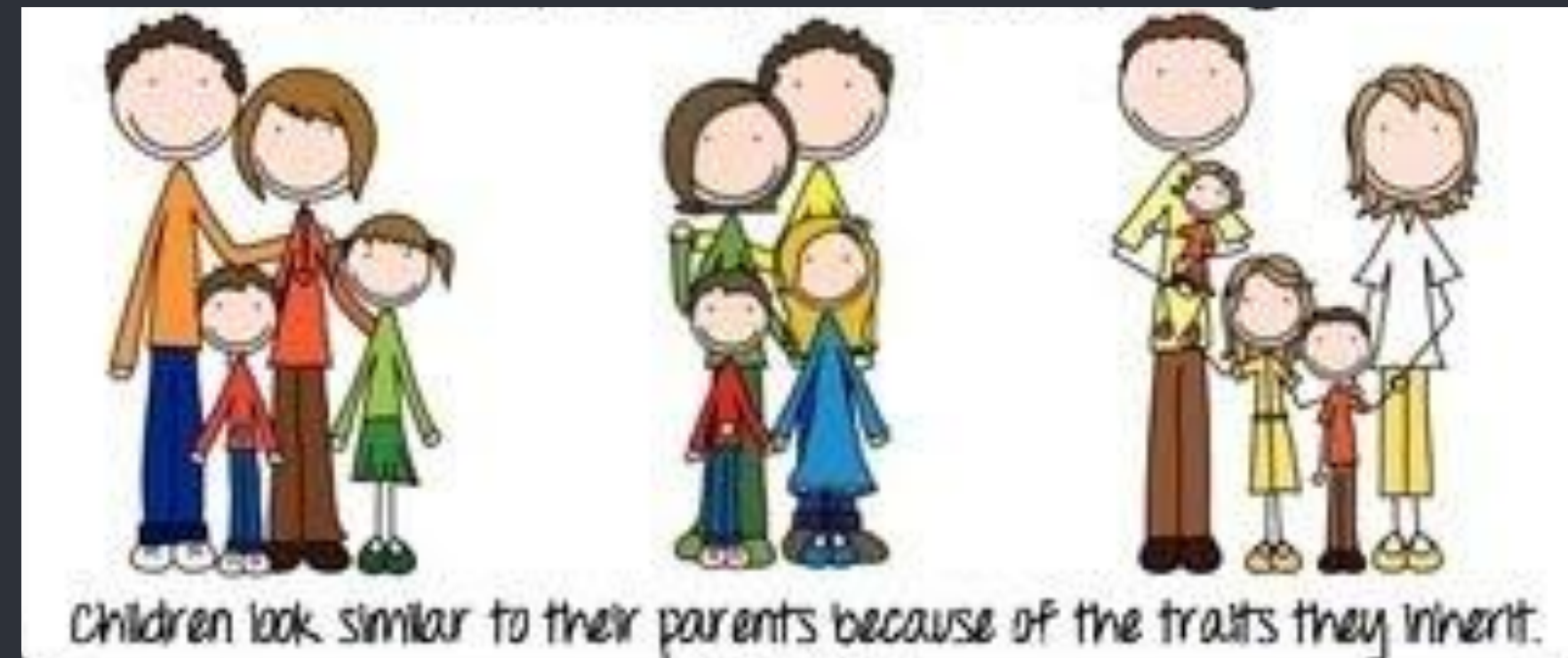
In animals including human beings, which reproduce sexually, we see distinct variations among individuals.

Variations more in sexual and less in asexual.



Heredity

The transfer of characters from parents to offspring is known as heredity.





Inheritance

The process through which characters pass from one generation to another is called inheritance.





Traits

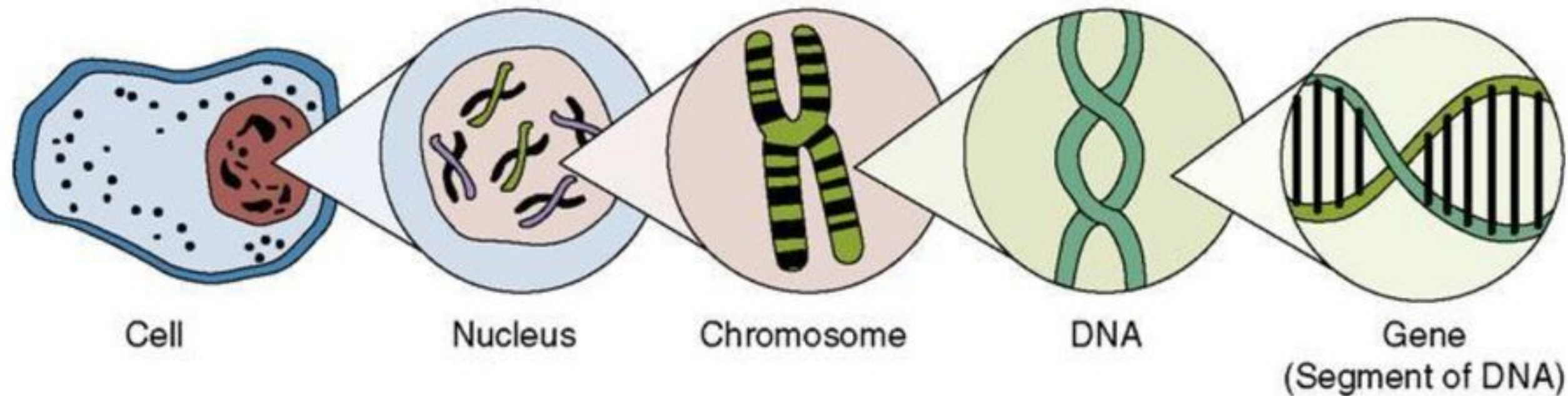
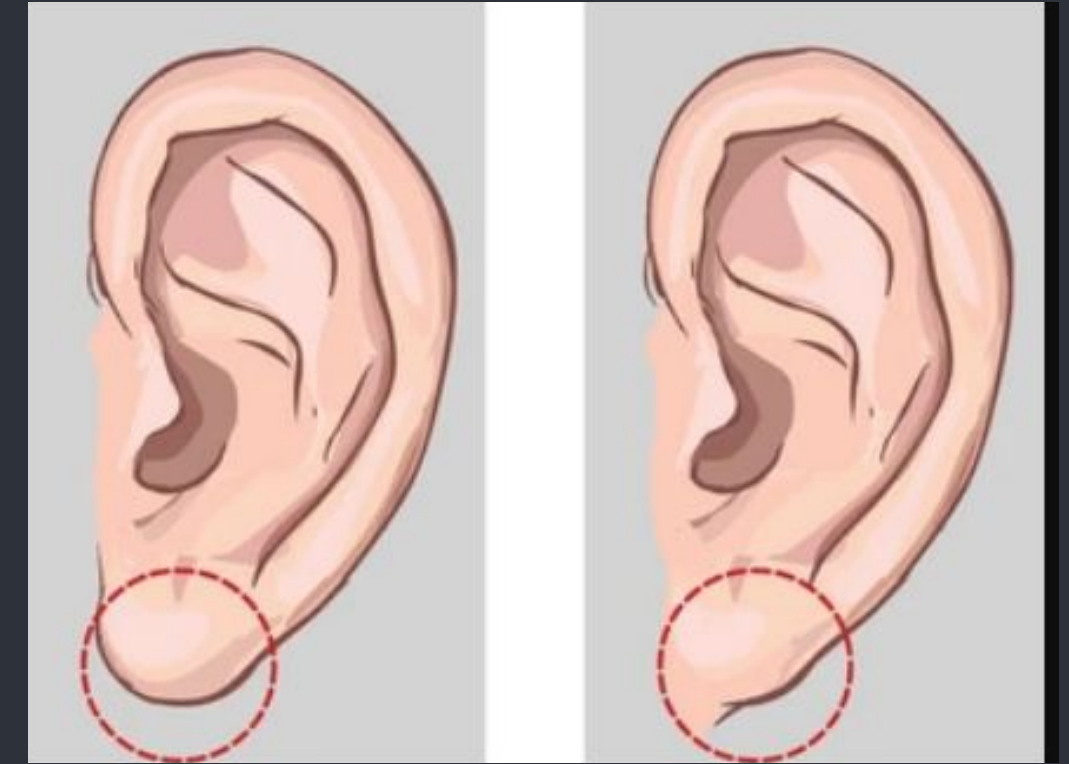
Traits are characteristic features of an organism, present in a physical form that is visible or in a physiological aspect of the organism.

	Acquired traits	Inherited traits
1.	Those traits that are developed by the individual during his lifetime.	Those traits which are present in an individual since birth.
2.	They are a result of changes in non-reproductive issues.	They are a result of changes in the DNA.
3.	They cannot be passed on to the progeny, e.g. pierced ear, large muscle size etc.	They are transmitted in the progeny, e.g. colour of eyes, skin or hair.

Genes



- Genes are the units of heredity which transfer characteristics from parents to their offsprings during reproduction.
- Due to the differences in genetic makeup, human population show a great deal of variations.
- It has been observed that attached and free earlobes are two variations found in human population.

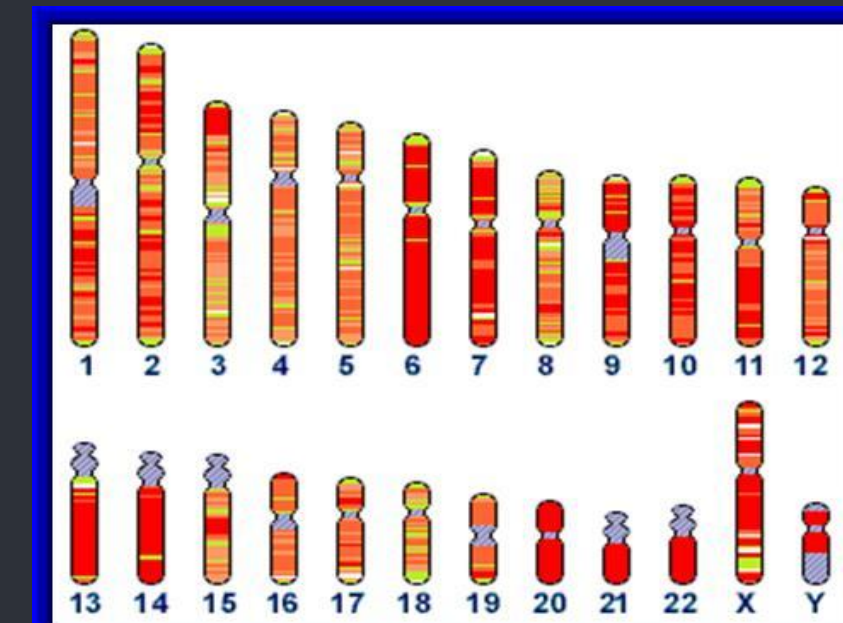
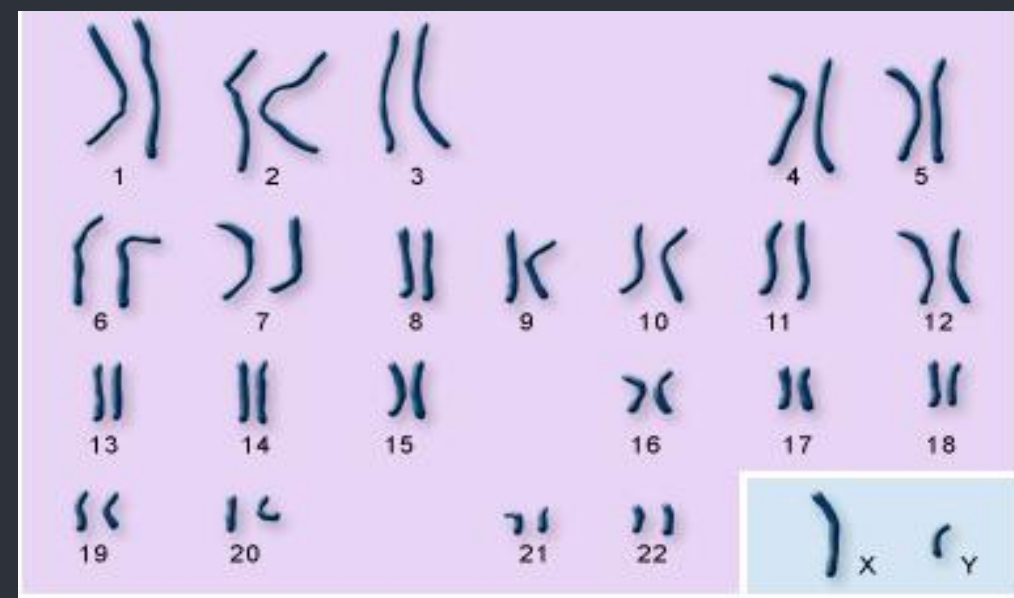
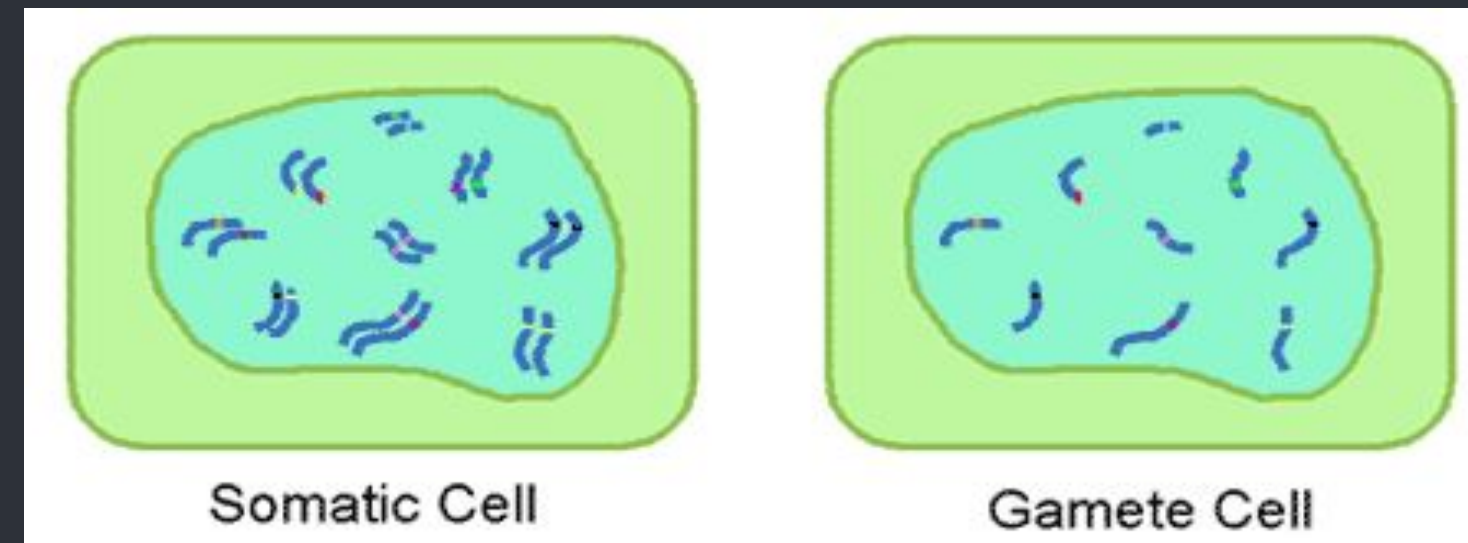


So we can say that genes are the functional units of DNA that define our characteristic.



Allele:

One of the different forms of a particular gene, occupying the same position on a chromosome.





Father of **GENETICS**

Why did Mendel choose pea plant?



- To study genetics, Mendel chose to work with pea plants because they have easily identifiable traits.

For example, pea plants are either tall or short, which is an easy trait to observe.

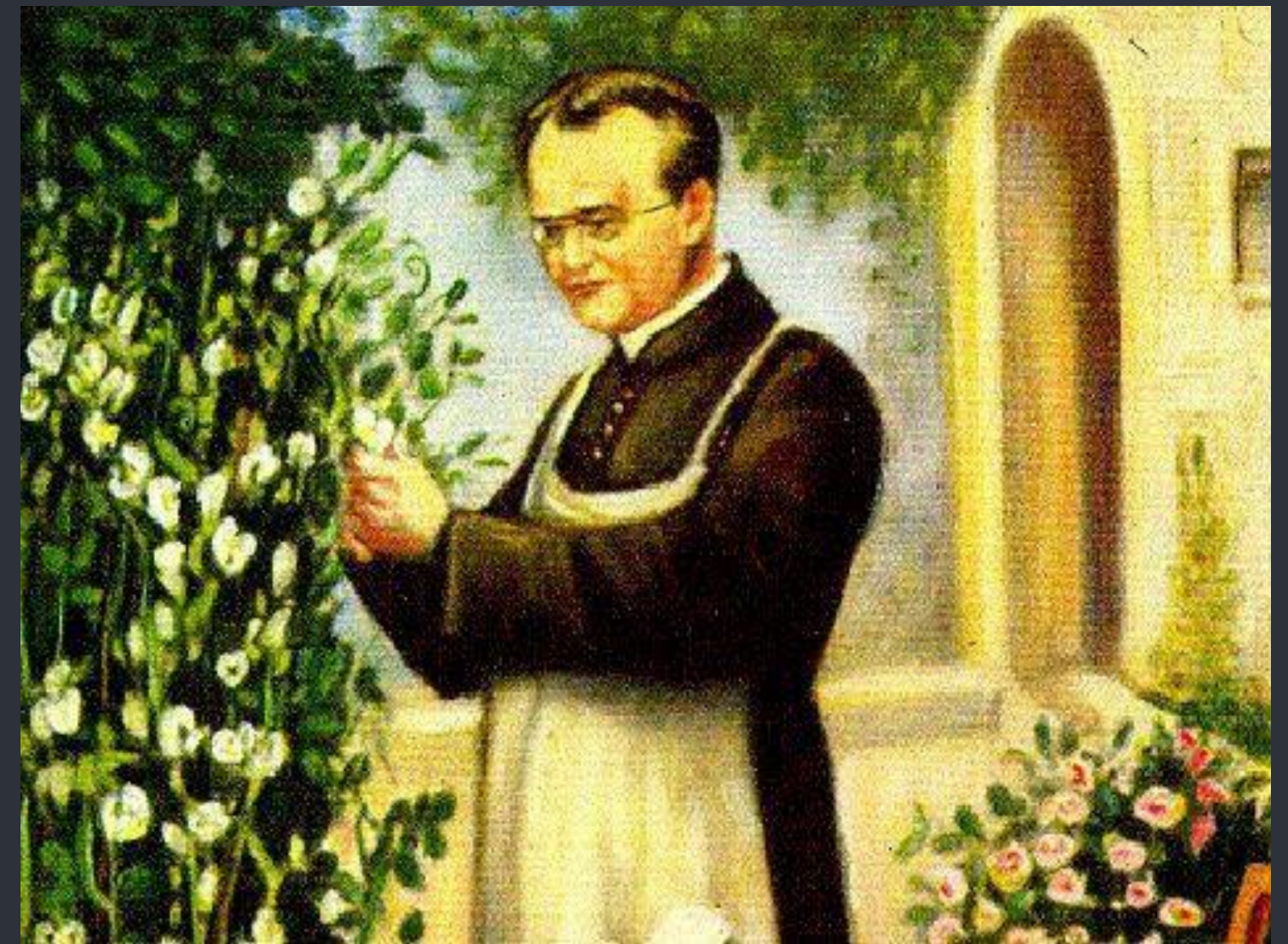
- Furthermore, pea plants grow quickly, so he could complete many experiments in a short period of time.
- Mendel also used pea plants because they can either self-pollinate or be cross-pollinated.



Why did Mendel choose pea plant?



- Self-pollination means that only one flower is involved; the flower's own pollen lands on the female sex organs.
- Since, Mendel could move pollen between plants, he could carefully control and then observe the results of crosses between two different types of plants.





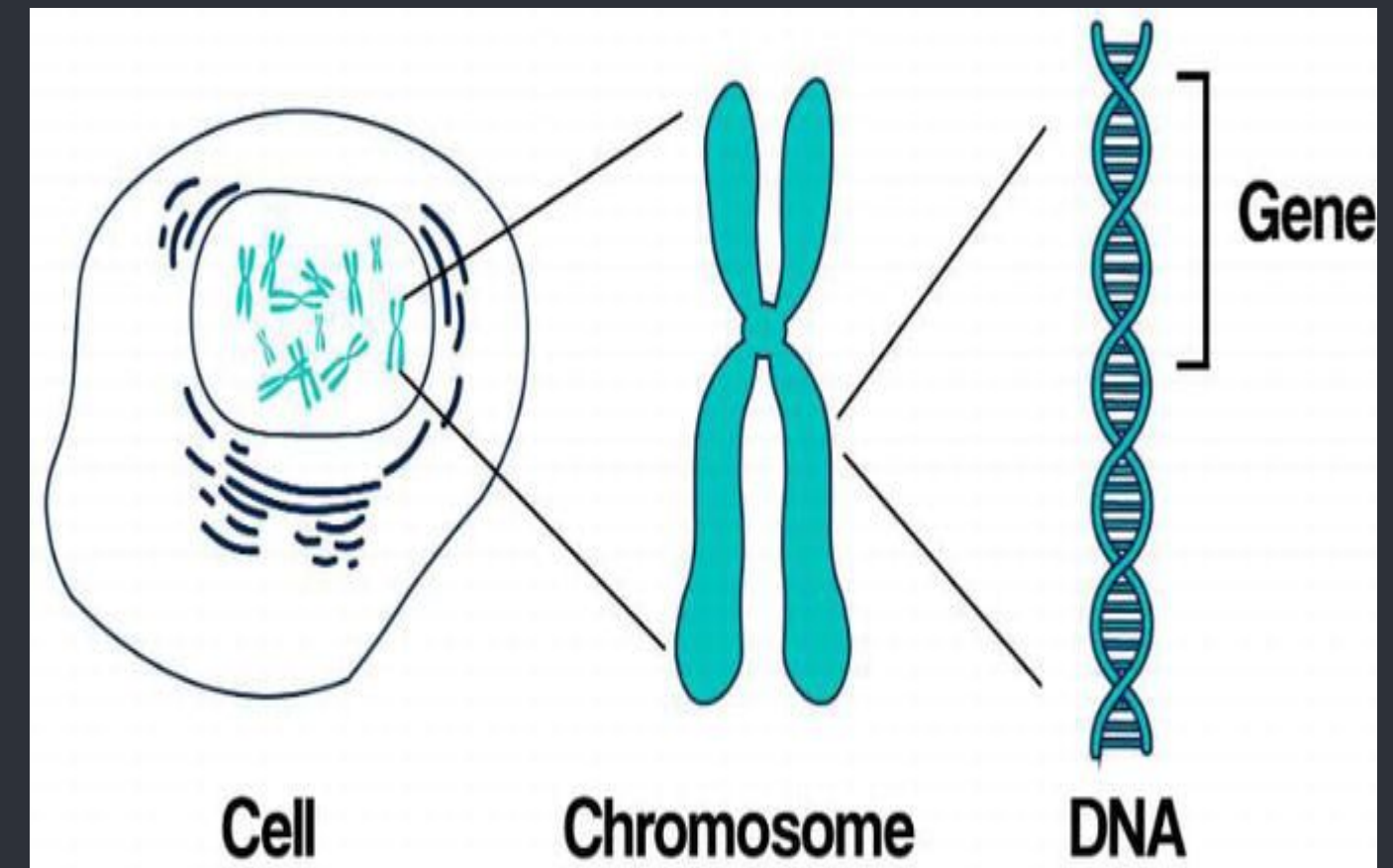
Character	Dominant Trait	×	Recessive Trait
Flower color	Purple 	×	White 
Flower position	Axial 	×	Terminal 
Seed color	Yellow 	×	Green 
Seed shape	Round 	×	Wrinkled 

Pod shape	Inflated 	×	Constricted 
Pod color	Green 	×	Yellow 
Stem length	Tall 	×	Dwarf 



Some Important terms to understand:

- **Chromosomes** : A thread-like structure in the nucleus of the cell. It appears during cell division and it carries genes.
- **Gene** : A functional unit of heredity. Present on chromosomes of cell nucleus. It is a piece of DNA that codes for one polynucleotide (protein).





Some Important terms to understand:

- **Allele:** DIFFERENT FORM OF A GIVEN GENE.

e.g. T-ALLELE responsible for tallness.

t- ALLELE responsible for dwarfness.

- **Phenotype:** It refers to observable physical appearance of a given organisms.

e.g. TALL PLANT, DWARF PLANT, ROUND SEEDS

- **Genotype:** It refers to genetic code responsible for any given phenotype.

e.g. (TT): is the genotype responsible for -----→ tallness of plant.

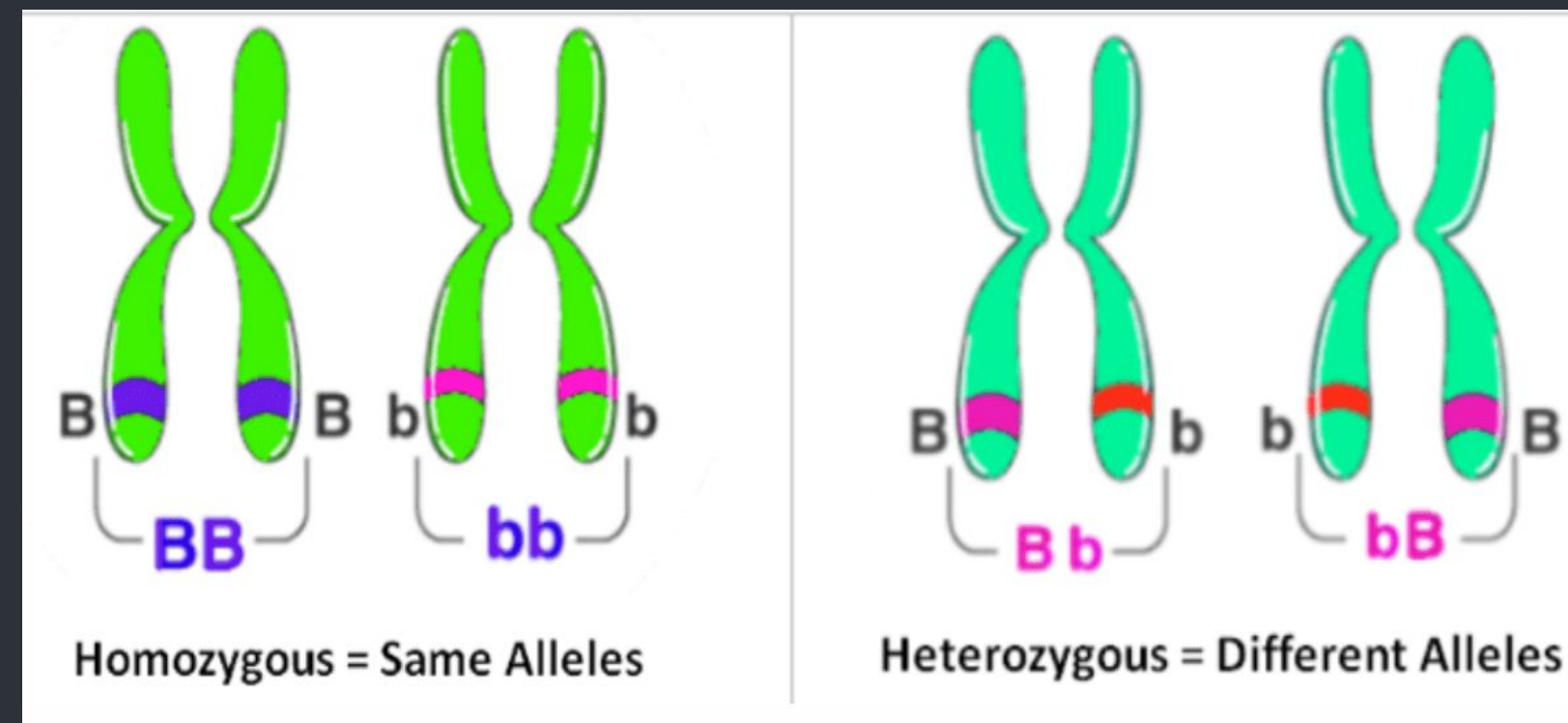
(tt): is the genotype responsible for-----→ dwarfness of plant.

(Tt): is the genotype responsible for---→tallness of plant.



Some Important terms to understand:

- **HOMOZYGOUS:** It refers to condition when SAME TYPE allele is present for a given phenotype. e.g. (TT) → homozygous tall
- **HETEROZYGOUS:** Different type of allele. e.g. (Tt) → heterozygous

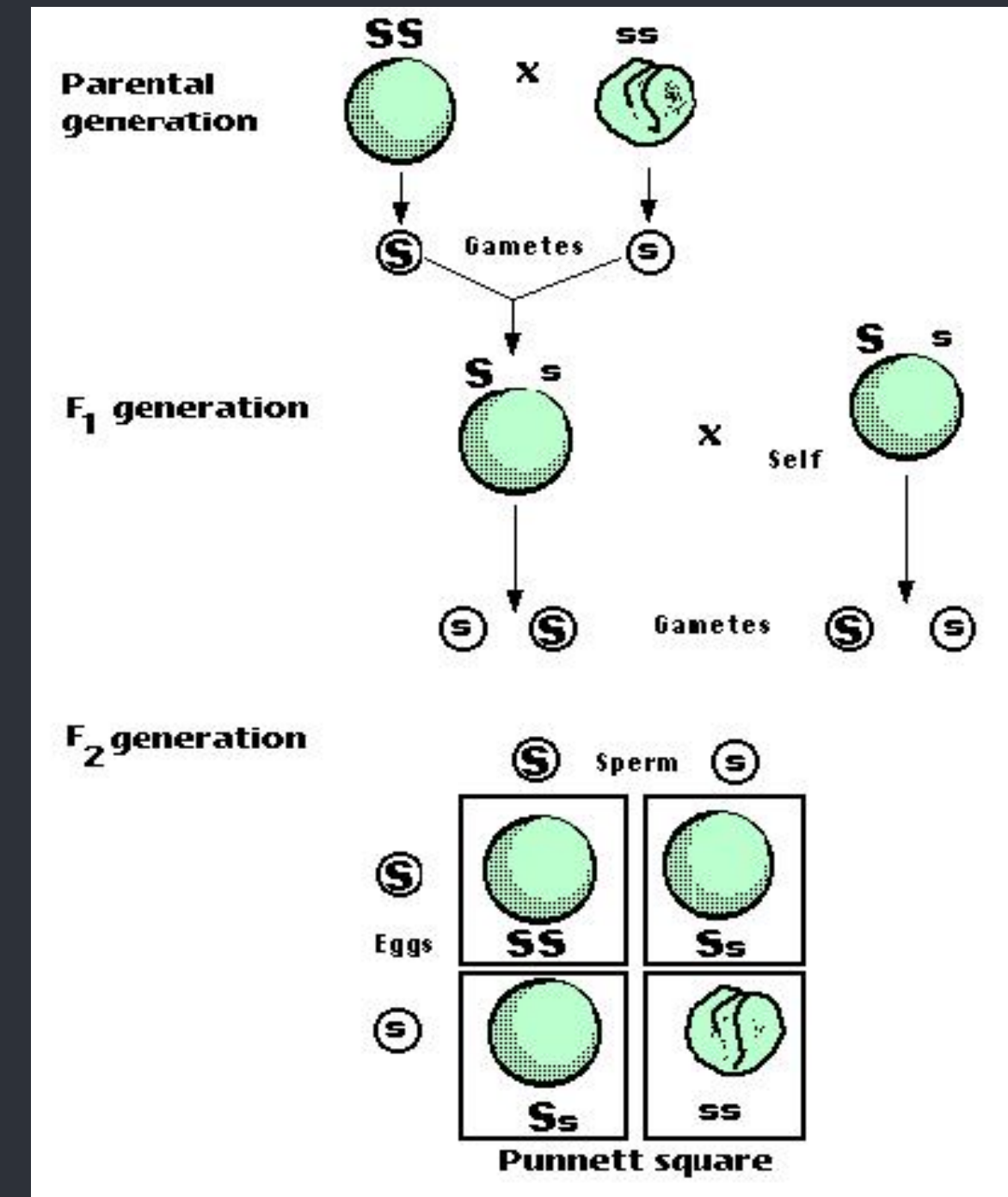




Some Important terms to understand:

- **F1 Generation** : Generation produced as a result of cross fertilization.
- **F2 Generation** : Generation produced as a result of selfing or self-fertilization of F1 generation.
- **Haploid cells** contain only one set of Chromosomes (n).
- **Diploid**, as the name indicates, contains two sets of chromosomes (2n)

ALL CELLS OF HUMAN BODY ARE DIPLOID EXCEPT SPERM AND OVUM





What did Mendel do?

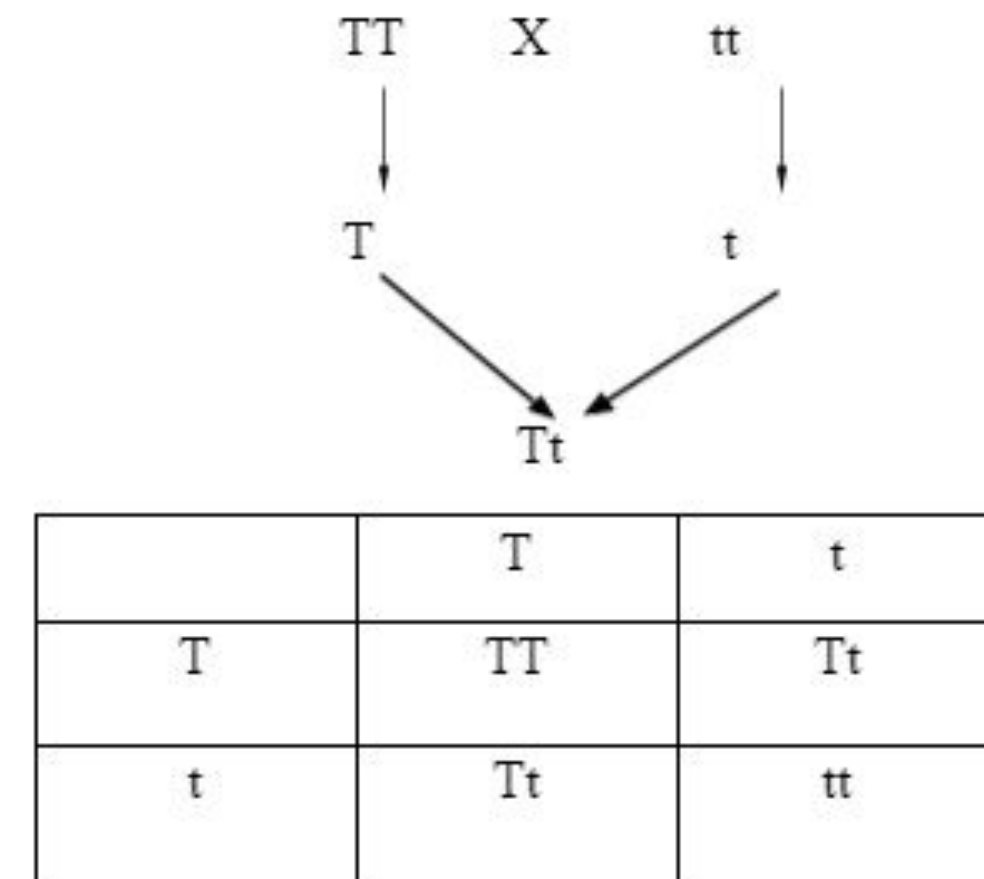
Mendel used a number of contrasting visible characters of garden peas – round/wrinkled seeds, tall/short plants, white/violet flowers and so on.

He took pea plants with different characteristics – a tall plant and a short plant, produced progeny by crossing them, and calculated the percentages of tall or short progeny.

Monohybrid Cross



- 3 : 1 is monohybrid phenotypic ratio.
- Genotype ratio = 1 : 2 : 1



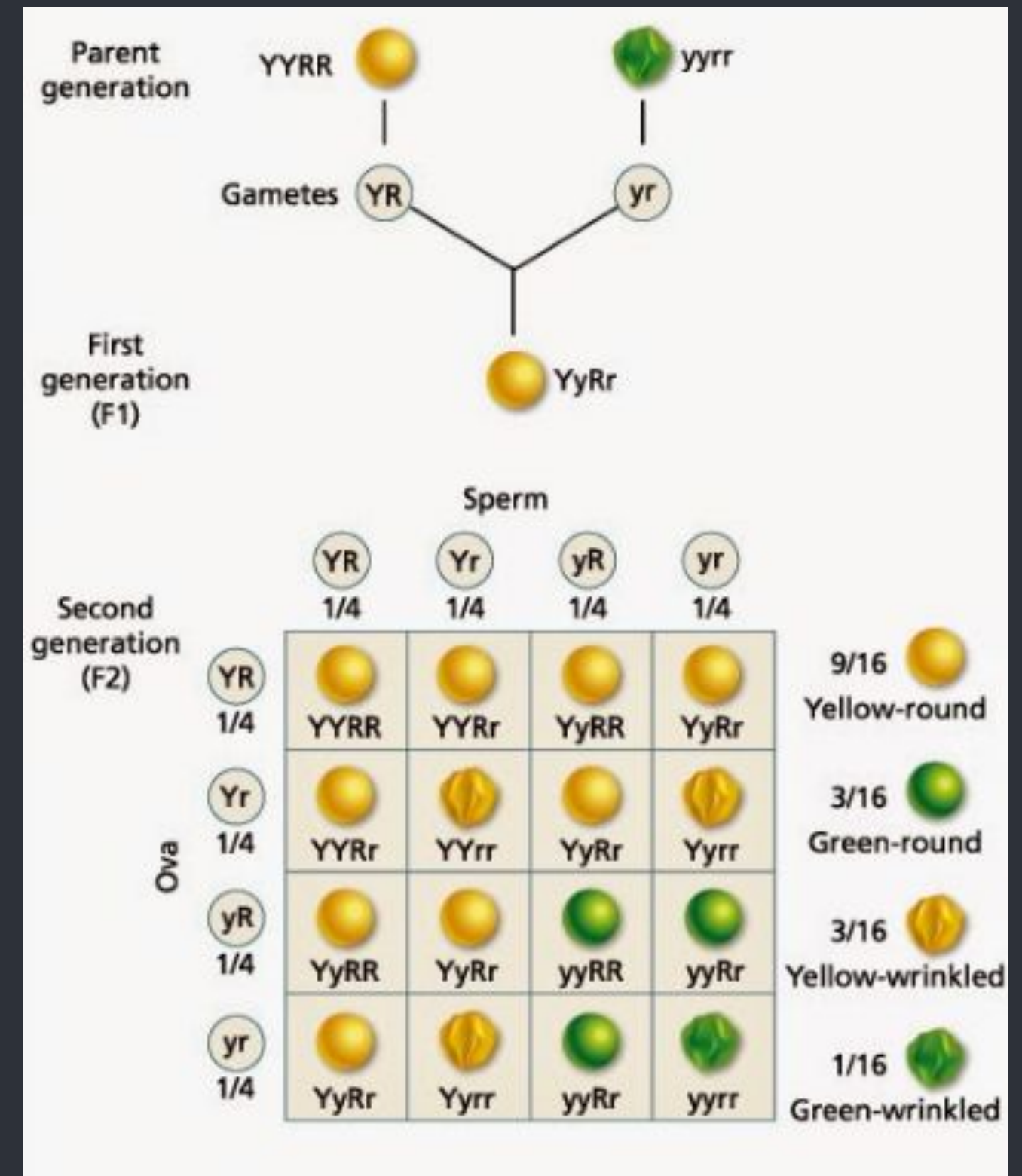
Phenotypic ratio: Tall: Short= 3:1

Genotypic Ratio: $TT:Tt:tt$ = 1:2:1

Dihybrid Cross



- Two characters are considered while crossing.
- A cross in which inheritance of two pairs of contrasting characters is simultaneously studied.
- Phenotypic Ratio (F1 Generation) :
9 : 3 : 3 : 1



Law of Dominance



Law of Dominance: Some alleles are dominant and cover up the recessive alleles.

- **First law of inheritance. In this law, each character is controlled by a pair of genes.**
- **If the pairs are heterozygous, one will always dominate the other.**
- **When parents with pure, contrasting traits are crossed together, only one form of trait appears in the next generation. The hybrid offspring will exhibit only the dominant trait in the phenotype.**

Law of Dominance



- In this explanation, both TT and Tt are tall plants, while only tt is a short plant. In other words, a single copy of 'T' is enough to make the plant tall, while both copies have to be 't' for the plant to be short.
- Traits like 'T' are called **dominant traits**, while those that behave like 't' are called **recessive traits**.

		Heterozygous Tall (Tt)	
		T	t
Heterozygous Tall (Tt)	T	TT homozygous tall 	Tt heterozygous tall 
	t	Tt heterozygous tall 	tt homozygous dwarf 

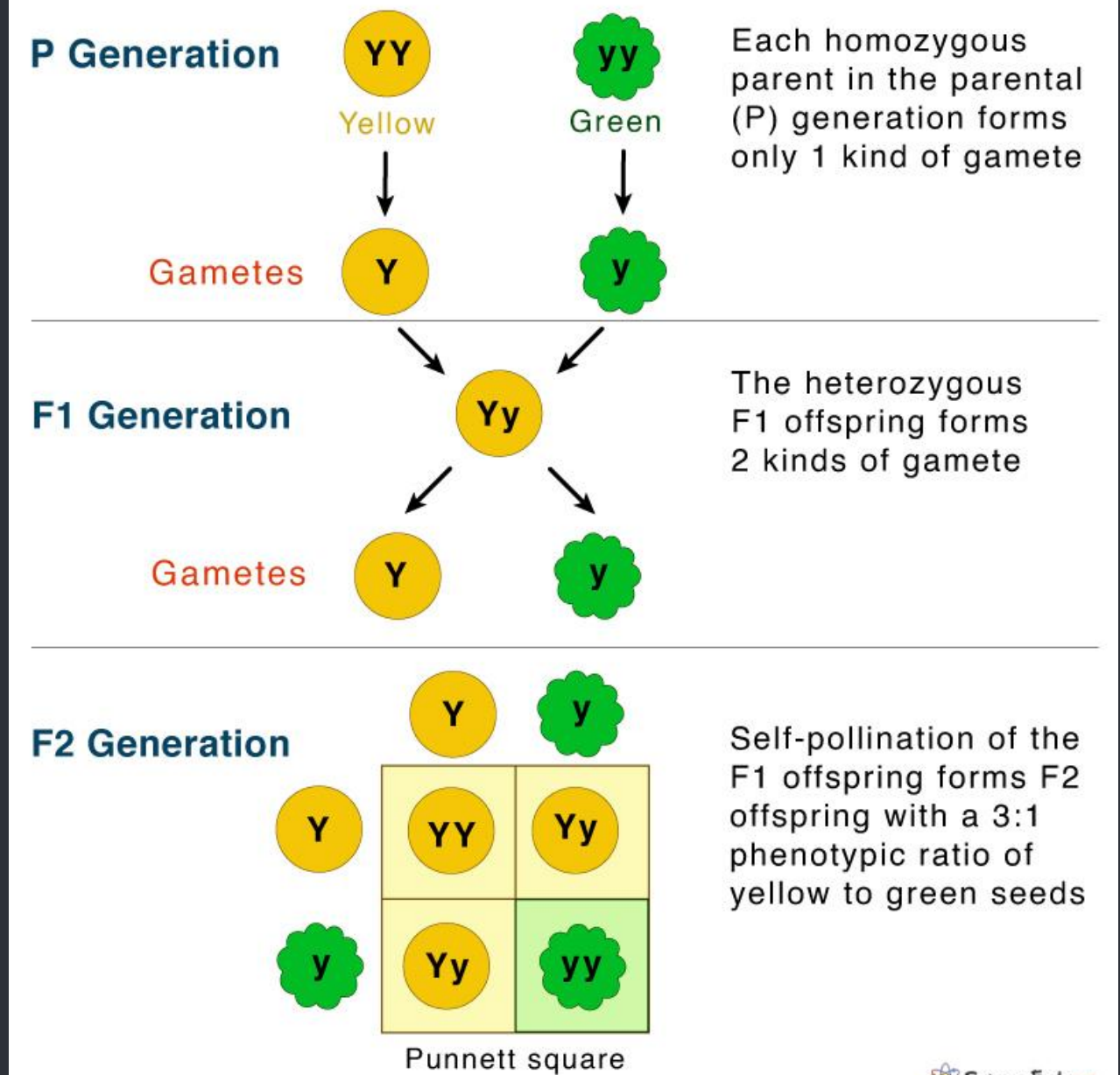
Law of Segregation



- **Second law of inheritance.**

An organism has two alleles for each gene but they can only pass on one.

Mendel's Law of Segregation

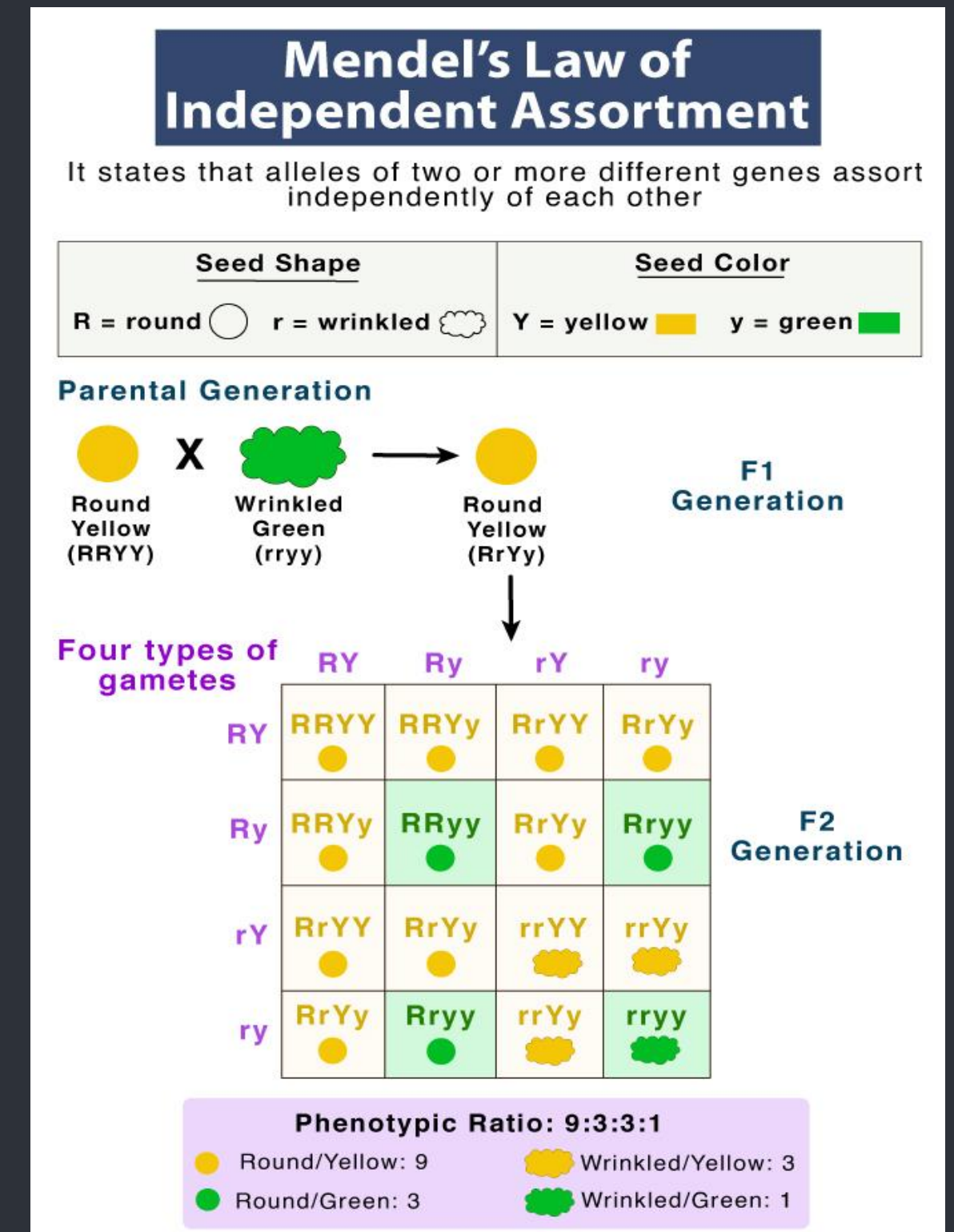


Law of Independent Assortment



It states that the:

- Assortment of each pair of traits is independent of the other.
- In other words, during gamete formation, one pair of trait segregates from another pair of traits independently. This gives each pair of characters a pair chance of expression.
- We can say that the allele of a gamete which is received for one gene does not influence the allele received for another gene.



How do traits get expressed?

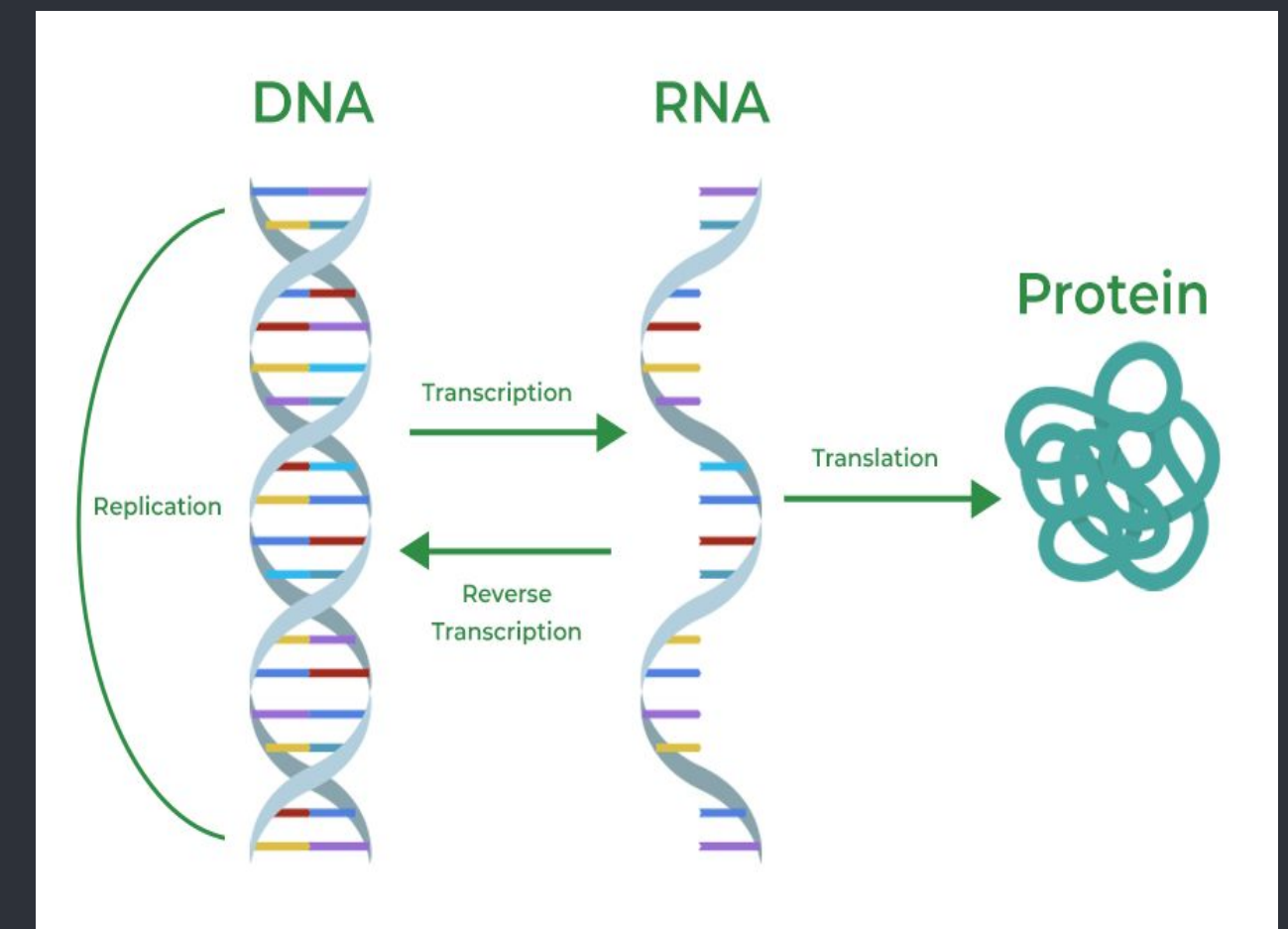


- Cellular DNA is the information source for making proteins in the cell.
- A part of DNA that provides information for one particular protein is called a gene for that protein. **For example:** the height of a plant depends upon the growth hormone which is in turn controlled by the gene.
- If the gene is efficient and more growth hormone is secreted the plant will grow tall. If the gene for that particular protein gets altered and less of it is secreted when the plant will remain short.

How do traits get expressed?



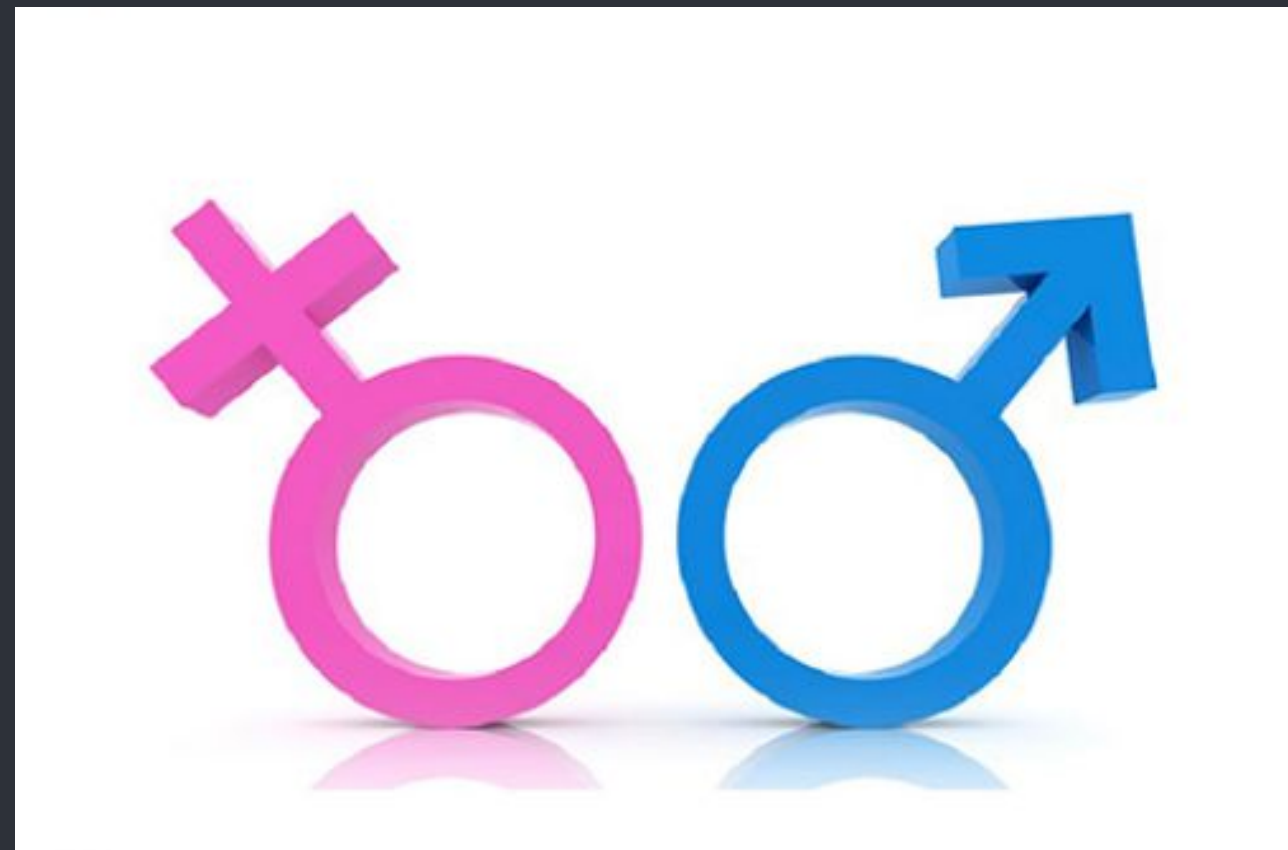
- Both the parents contribute equally to the DNA of next generation during sexual reproduction.
- They actually contribute a copy of the same gene for example: when tall plant is crossed with short plant the gametes will have single gene either for tallness or for shortness. F1 generation will get one gene for tallness and other for shortness also.



Sex Determination



The process by which sex of an individual is decided or determined based on its genetic material composition is known as SEX DETERMINATION.





How is the sex of a newborn individual determined?



Sex determination in Humans



22 PAIRS : Autosomes

1 PAIR : Sex Chromosomes

Male - XY

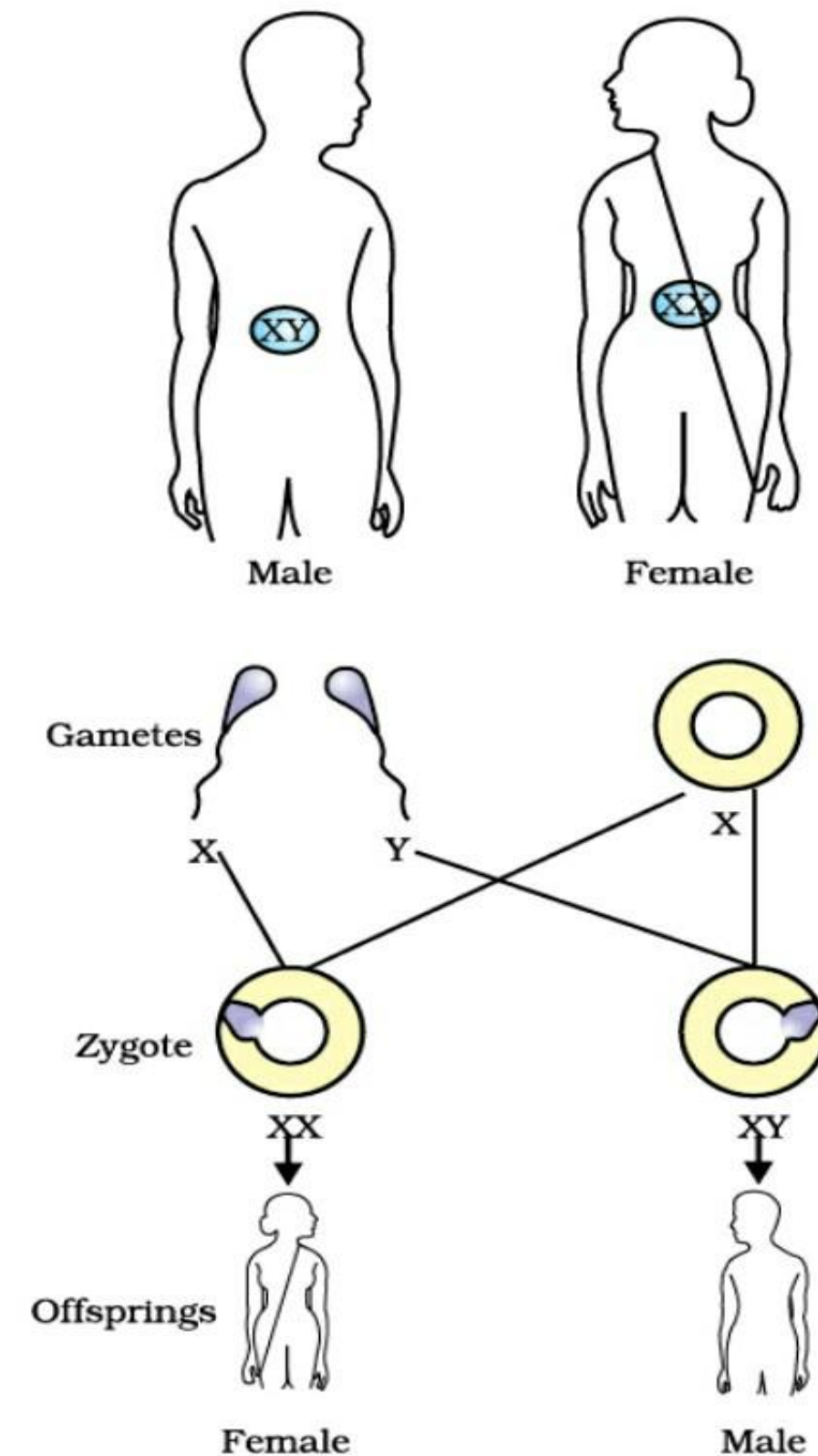
Female - XX

CHROMOSOMES	AUTOSOMES
Chromosomes that are not really involved in determining sex	Chromosomes that determine sex and sex-related hormonal traits.
Males and females have similar traits	Males (XY) and females have (XX) which is dissimilar
In humans there are usually 44 or 22 pairs	In humans only one pair is present

Sex determination in Humans

All children will inherit an X chromosome from their mother regardless of whether they are boys or girls.

Thus, the sex of the children will be determined by what they inherit from their father.





IMPORTANT QUESTIONS





1. What is the Dihybrid ratio?

A. 9:3:3:1

B. 9:3:3:2

C. 9:3

D. 3:1



2. If a round, green seeded pea plant ($RR\ yy$) is crossed with wrinkled, yellow seeded pea plant ($rr\ YY$), the seeds produced in F1 generation are:

- A. Round and yellow**
- B. Round and green**
- C. Wrinkled and green**
- D. Wrinkled and yellow**



3. What is the difference between alleles and genes?

Gene	Allele
A gene is a portion of DNA that determines a certain trait.	An allele is a specific form of a gene.
Genes are responsible for the expression of traits.	Alleles are responsible for the variations in which a given trait can be expressed.
Genes do not occur in pairs.	Alleles occur in pairs.
Eye color, hair color, hairline shape	Blue eyes, blonde hair, V-shaped hairline



4. What is the difference between inherited and acquired traits?

	Acquired traits	Inherited traits
1.	Those traits that are developed by the individual during his lifetime.	Those traits which are present in an individual since birth.
2.	They are a result of changes in non-reproductive issues.	They are a result of changes in the DNA.
3.	They cannot be passed on to the progeny, e.g. pierced ear, large muscle size etc.	They are transmitted in the progeny, e.g. colour of eyes, skin or hair.



5. How do Mendel's experiments show that traits may be dominant or recessive?

Mendel selected true breeding tall (TT) and dwarf (tt) pea plants. Then, he crossed these two plants. The seeds formed after fertilisation were grown and these plants that were formed represent the F1 generation. All the F1 plants obtained were tall. Then, Mendel self-pollinated the F1 plants and observed that all plants obtained in the F2 generation were not tall. Instead, 1/4th of the F2 plants were short. From this experiment, Mendel concluded that the F1 tall plants were not true breeding. They were carrying traits of both short height and tall height. They appeared tall only because the tall trait is dominant over the dwarf trait.



6. A blue colour flower plant denoted by BB is crossbred with a white colour flower plant denoted by bb.

(a) State the colour of flower you expect in their F₁ generation plants.

(b) What must be the percentage of white flower plants in F₂ generation if flowers of F₁ plants are self-pollinated?

(c) State the expected ratio of the genotypes BB and Bb in the F₂ progeny.

